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	MSDS for HP-NanoLite

**MATERIAL SAFETY DATA SHEET (MSDS)
(HP-NanoLite)**



Section 1. Chemical Product and Company Identification

Product Name	HP-NanoLite
Product No.	NA
Application	Anti-friction, Anti-wear and Extreme Pressure additive for Grease
Supplier	HPCL R&D Centre, Bangalore
Emergency Telephone	080-28078658/8680

Identified uses: Used as friction modifier, wear resistance and extreme pressure, improving load bearing capacity of the greases.
HPCL R&D product.

Section 2. Hazards Identification

Hazard Identification:

Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008

Skin irritation (Category 2), H315

Label elements:

Labelling according Regulation (EC) No 1272/2008

Irritation of the respiratory system by inhalation, bluish color staining in contact with skin and if swallowed

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Hazard statement(s)

H317 May cause an allergic skin reaction

Precautionary statement(s):

P261 Avoid breathing dust
P272 Contaminated work clothing should not be allowed out of the workplace
P280 Wear Protective gloves
P302+P353 IG ON SKIN: Wash with plenty of water
P333+P313 If skin irritation or rash occurs: Get medical advice/attention
P362+P364 Take off contaminated clothing and wash it before reuse.

Supplemental Hazard statements: None

Other Hazards:

This substance/mixture contains no components considered to be either persistent, bio-accumulative and toxic (PBT), or very persistent and very bio-accumulative (vPvB) at levels of 0.1% or higher.

Section 3. First Aid Measures

Description of first aid measures:

General advice: ***Always seek professional medical attention after first aid measures are provided.***
Show this safety data sheet to the doctor in attendance.

If inhaled: If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.

In case of skin contact: Wash off with soap and plenty of water (at least 15-20mins). Consult a physician.

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In case of eye contact: Flush eyes with water as a precaution until no evidence of chemical remains. Consult a physician.

If swallowed: Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

Most important symptoms and effects, both acute and delayed
The most important known symptoms and effects are described in the labelling

Indication of any immediate medical attention and special treatment needed: No data available.

Section 4. Composition and Information on Ingredients

Characteristics	Unit	Standard Request	Test Method
Appearance	-	White Powder	Visual
Zr(HPO ₄) ₂ ·H ₂ O content	% (w/w)	99 min	HPCL
Zr	% (w/w)	38-40	XRF (HPCL)
P	% (w/w)	17-20	XRF (HPCL)
Water insoluble	% (w/w)	99.9 max	HPCL
Loss of Moisture	% (w/w)	5 max	(TGA (HPCL)
Average Particle Size (thickness)	nm	200-400	FESEM (HPCL)
pH value	-	7	HPCL
Density	g/cm ³	1.09	Tapping mode (HPCL)

1) Delivery specification: The product is monitored on a regular basis to ensure that it adheres to the specified values.

2) Technical data: The technical data are used solely to describe the product and are not subject to regular monitoring

Section 5: Firefighting measures **Extinguishing media**

Suitable extinguishing media:

Regular dry chemical, carbon dioxide, water, regular foam. Use regular foam or flood with fine water spray in case of large fire.

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Unsuitable extinguishing media:

For this substance/mixture no limitations of extinguishing agents are given.

Special hazards arising from the substance or mixture

Oxides of phosphorus

Zirconium metal oxides

Combustible.

Development of hazardous combustion gases or vapours possible in the event of fire.

Advice for firefighters

Stay in danger area only with self-contained breathing apparatus. Prevent skin contact by keeping a safe distance or by wearing suitable protective clothing.

Further information

Suppress (knock down) gases/vapors/mists with a water spray jet. Prevent fire extinguishing water from contaminating surface water or the ground water system.

Section 6: Accidental release measures

Personal precautions, protective equipment and emergency procedures

Advice for non-emergency personnel: Avoid inhalation of dusts. Avoid substance contact.

Ensure adequate ventilation. Evacuate the danger area, observe emergency procedures, consult an expert.

Environmental precautions

Do not let product enter drains.

Methods and materials for containment and cleaning up

Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions. Take up dry. Dispose of properly. Clean up affected area. Avoid generation of dusts.

Section 7: Handling and storage

Precautions for safe handling:

Avoid contact with skin and eyes. Avoid inhalation of vapor or mist.

Keep away from sources of ignition - No smoking. Take measures to prevent the buildup of electrostatic charge.

Conditions for safe storage, including any incompatibilities:

Store in a cool, dry, ventilated area away from source of heat. Store in a tightly closed container.

Keep separated from incompatible substances. Direct sunlight should be avoided.

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Storage class (TRGS 510): Combustible Solids

Specific end use(s): Apart from the uses mentioned in section 1 no other specific uses are stipulated

Section 8: Exposure controls/personal protection

Control parameters	Ingredients with workplace control parameters
Exposure controls:	Appropriate engineering controls

Eye/face protection

Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU). Safety glasses

Skin protection

This recommendation applies only to the product stated in the safety data sheet, supplied by us and for the designated use. When dissolving in or mixing with other substances and under conditions deviating from those stated in EN374 please contact the supplier of CE-approved gloves

Body Protection

Wear appropriate chemical resistant clothing, including boots, gloves, apron as appropriate to skin contact.

Respiratory protection

Required when dusts are generated.

Eye Protection

Wear splash resistant safety goggles with a face shield. Provide an emergency eye wash fountain and quick shower in the immediate work area.

Section 9: Physical and chemical properties

Information on basic physical and chemical properties

- | | |
|------------------------|--------------------|
| a) Physical State | Power |
| b) Appearance | White |
| c) Odor | NA |
| d) pH | ~7 |
| e) Boiling Point | NA |
| f) Freezing Point | NA |
| g) Specific Gravity | NA |
| h) Solubility in water | Not-Soluble |

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Section 10: Stability and reactivity

Reactivity

The following applies in general to flammable organic substances and mixtures: in correspondingly fine distribution, when whirled up a dust explosion potential may generally be assumed.

Chemical stability

The product is chemically stable under standard ambient conditions (room temperature) .

Possibility of hazardous reactions

Violent reactions possible with: Strong oxidizing agents

Conditions to avoid no information available

Incompatible materials No data available

Hazardous decomposition products: In the event of fire: see section 5

Section 11: Toxicological information

Oral Toxicity Information is not available for this product.

Dermal Toxicity No information is available for this compound.

Inhalation No information is available for this compound.

Irritation Information for this product is not available.

Section 12: Ecological information

Toxicity: No data available

Persistence and degradability: No data available

Bio accumulative potential: No data available

Mobility in soil: No data available

Section 13: Disposal considerations

Waste treatment methods:

Product: Dispose all waste materials including liquid, contaminated absorbent and material from spill clean-up procedures in accordance with all applicable government regulations.

Section 14: Transport information

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Land Transport ADR/RID No Classification assigned.

Air Transport IATA/CAO No Classification assigned.

Maritime Transport IMDG No Classification assigned.

Section 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

This material safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.

Authorisations and/or restrictions on use

Other regulations

Take note of Dir 94/33/EC on the protection of young people at work.

15.2 Chemical Safety Assessment

For this product a chemical safety assessment was not carried out

Risk of serious damage to eyes. In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Wear eye / face protection equipment's.

Section 16: Other information

The data given here is based on current knowledge and experience. The purpose of this safety Data Sheet is to describe the product in terms of the Safety requirements. The data does not signify any warranty with regard to the products' properties

Further information

The information in the MSDS is based on data which is considered to be accurate. However, we makes no guarantees or warranty, either expressed or implied on the accuracy or completeness of this information. The conditions of handling, storage, use and disposal are beyond our control and

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